Notes of HDP[1]

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August 3, 2026

[1]Based on IEOR8100 High-dimensional Probability with Applications at Columbia University, taught by Prof. Kaizheng Wang

Outline

Outline

Intro

Informal version of

- Concentration
- Suprema

Preliminaries and Notations

- $a \lesssim b$: $a \leq Cb$ where C is some absolute constant.
- -[Jensen inequality] for any random variable X and a convex function $f : \mathbb{R} \to \mathbb{R}$, we have

$$f(\mathbb{E}\, X) \leq \mathbb{E}\, f(X).$$

  More generally, this holds for any random vector X taking values in $\mathbb{R}^n$ and any convex function $f : \mathbb{R}^n \to \mathbb{R}$.

Outline

Intro - Concentration inequalities

(1) Variance version of concentration

$$\mathrm{Var}(f) \lesssim \sum_{i=1}^{n} \mathbb{E}\left[(D_i f)^2\right]$$

the variance of a function f of independent (or weakly dependent) random variables is small if the "gradient" of f is small.

(2) Sub-Gaussianity, a sharper control on the distribution of the fluctuations

$$\|f - \mathbb{E}f\|_{\psi_2}^2 \lesssim \left\| \sum_{i=1}^{n} D_i f \right\|_{\infty}^2 \quad \text{by simple martingale}$$

$$\|f - \mathbb{E}f\|_{\psi_2}^2 \lesssim \left\| \sum_{i=1}^{n} D_i f \right\|_{\infty}^2 \quad \text{by entropy method}$$

-

(3)

'We will repeatedly encounter this idea in the setting of concentration inequalities: typically the expectation of the "square gradient" controls the variance, while a uniform bound on the "square gradient" controls the subgaussian variance proxy.'

Outline

Tensorization property of the Variance

Efron-Stein

- relationship between the variance of the overall function and the sum of the individual expected variance[] contributions

[Efron-Stein] Let $X_1, \ldots, X_n$ be independent random variables, define

$$\mathrm{Var}_{i} f(x_1, \ldots, x_n) := \mathrm{Var}[f(x_1, \ldots, x_{i-1}, X_i, x_{i+1}, \ldots, x_n)] = \mathrm{Var}(f \mid X_{-i})$$

Then

$$\mathrm{Var}[f(X_1, \ldots, X_n)] \leq \sum_{i=1}^{n} \mathbb{E}\left[\mathrm{Var}_i f(X_1, \ldots, X_n)\right]$$

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proof: martingale method

Bounded Difference Inequalities

Variance version

- relationship between variance of overall function and the sum of Difference $D_i f$ with respect to individual variable

[Bounded Difference Inequalities] Let $X_1, \ldots, X_n$ be independent random variables, define

$$D_i f(x_1, \ldots, x_n) = \sup_{x_i} f - \inf_{x_i} f$$

$$D_i^- f(x_1, \ldots, x_n) = f - \inf_{x_i} f$$

Then

$$\mathrm{Var}(f) \leq \frac{1}{4} \sum_{i=1}^{n} \mathbb{E}\left[(D_i f)^2\right]$$

$$\mathrm{Var}(f) \leq \frac{1}{4} \sum_{i=1}^{n} \mathbb{E}\left[(D_i^- f)^2\right]$$

Example:

$D_i f$, a 'discrete derivative' of the function with respect to the i-th variable

Poincaré Inequalities

- from 'variance/difference' to 'derivative'

$$\mathrm{Var}(f) \lesssim \mathbb{E}\left[\|\mathrm{gradient}(f)\|^2\right]$$

[Convex Poincaré Inequalities] Let $\{X_i\}_{i=1}^n$ be independent random variables, $X_i \in [a, b]$, $f : \mathbb{R}^n \to \mathbb{R}$ convex and differentiable. Then

$$\mathrm{Var}(f) \leq (b-a)^2 \cdot \mathbb{E}\,\|\nabla f(X_1, \ldots, X_n)\|_2^2$$

---

proof: first relate 'difference' with 'derivative' with respect to certain assumption on X and f, then use efron-stein/bounded difference inequalities

Poincaré Inequalities

General version

Poincaré Inequalities

Application to Sampling

Outline

Intro

Subgaussian Concentration

- Subgaussianity: another way to describe concentration
- Entropy Method
- Log-Sobolev Inequalities

Tail probability bounded by 'Moments'

Chebyshev inequality, Chernoff bound

Let X be a random variable. Then, for any $t > 0$, we have

- -[Chebyshev inequality]

$$\mathbb{P}\{|X - \mathbb{E}X| \geq t\} \leq \frac{\mathbb{E}\,|X - \mathbb{E}X|^p}{t^p}$$

- -[Chernoff bound, exponential moment method] Define the log-moment generating function ψ of a random variable X and its Legendre dual ψ^* as

$$\psi(\lambda) := \log \mathbb{E} e^{\lambda(X - \mathbb{E}X)}, \qquad \psi^*(t) = \sup_{\lambda \geq 0}\{\lambda t - \psi(\lambda)\}.$$

Then

$$\mathbb{P}\{X - \mathbb{E}X \geq t\} \leq e^{-\psi^*(t)} \quad \text{for all } t \geq 0.$$

bounding tail probability $\rightarrow$ bounding the MGF.

proof: by Markov Inequality on nonnegative increasing function of $X - \mathbb{E}X$

Subgaussian Variable

[Subgaussian properties] Let X be a random variable. The follows are equivalent, with $K_i > 0$ differing by at most an absolute constant.

(i) (Tail bound) There exists $K_1 > 0$ such that

$$\mathbb{P}\{|X| \geq t\} \leq 2\exp(-t^2/K_1^2) \quad \text{for all } t \geq 0.$$

(ii) (Moment growth bound) There exists $K_2 > 0$ such that

$$\frac{\|X\|_{L^p}}{\sqrt{p}} = \frac{(\mathbb{E}|X|^p)^{1/p}}{\sqrt{p}} \leq K_2 \quad \text{for all } p \geq 1.$$

[Subgaussian norm] $\|X\|_{\psi_2} = \sup_{p\geq 1}\left\{\frac{(\mathbb{E}|X|^p)^{1/p}}{\sqrt{p}}\right\}$, the smallest K_2.

(iii) (Finite MGF of X^2) There exists $K_3 > 0$ such that

$$\mathbb{E}\exp(X^2/K_3^2) \leq 2.$$

Moreover, if $\mathbb{E}X = 0$, (i)–(iii) are equivalent to the following one:

(iv) (MGF bound) There exists $K_4 > 0$ such that

$$\mathbb{E}\exp(\lambda X) \leq \exp(\lambda^2 K_4^2) \quad \text{for all } \lambda \in \mathbb{R}.$$

[Centering] Any subgaussian random variable X satisfies

$$\|X - \mathbb{E}X\|_{\psi_2} \leq C\|X\|_{\psi_2}.$$

Subgaussianity

a centered version

[Subgaussianity] A random variable X is σ^2-subgaussian if

$$\mathbb{E}e^{\lambda(X-\mathbb{E}X)} \leq e^{\lambda^2\sigma^2/2}, \quad \forall\lambda \in \mathbb{R}$$

If X is σ^2-subgaussian, then

$$\mathbb{P}\{X - \mathbb{E}X \geq t\} \leq e^{-t^2/2\sigma^2}, \quad \mathbb{P}\{X - \mathbb{E}X \leq -t\} \leq e^{-t^2/2\sigma^2}.$$

This is also equivalent to

$$\|X - \mathbb{E}X\|_{\psi_2}^2 \lesssim \sigma^2.$$

σ^2 is called the variance proxy because it shows like variance, the sum of n independent σ^2-subg variable is $\frac{1}{n}\sigma^2$-subg.

Subgaussianity

What r.v.'s are subgaussian?

- Gaussian
- A bounded r.v. [Hoeffding] If $a \leq X \leq b$ a.s. for constants a & b, then X is $\frac{(b-a)^2}{4}$-subg.

Subgaussianity of a function

Bounded Difference Inequalities

Subgaussianity is another description of concentration, and now we discuss the subgaussianity of function $f(X_1, \dots, X_n)$.
[Azuma-Hoeffding inequality] (concentration of the sums of martingale differences)

[McDiarmid] If $\{X_i\}_{i=1}^n$ are indep, $\|D_i f\|_\infty < \infty$ for all i, then $f(X_1, \dots, X_n)$ is subgaussian with variance proxy $\frac{1}{4}\sum_{i=1}^n \|D_i f\|_\infty^2$

$$\mathbb{P}\{f - \mathbb{E}f \geq t\} \leq e^{-2t^2/\sum_{i=1}^n \|D_i f\|_\infty^2}$$

$$\|f - \mathbb{E}f\|_{\psi_2}^2 \lesssim \sum_{i=1}^n \|D_i f\|_\infty^2.$$

Actually subgaussian property does not tensorize naturally, and the martingale method can only partially address this issue. Next, we explore the entropy method to derive a better bound.

Entropy and Subgaussian

[Entropy] The entropy of a nonnegative random variable Z is

$$\mathrm{Ent}[Z] := \mathbb{E}[Z \log Z] - \mathbb{E}[Z] \log \mathbb{E}[Z].$$

A sufficient condition for subgaussian in terms of entropy:
[Herbst] If $\mathrm{Ent}(e^{\lambda X}) \leq \frac{\lambda^2\sigma^2}{2}\mathbb{E}e^{\lambda X}$, $\forall \lambda$, then X is σ^2-subg.

Tensorization property of the Entropy

Let $X_1, \ldots, X_n$ be independent random variables, $f \geq 0$, define

$$g_i(x_1, \ldots, x_n) = \mathrm{Ent}\left[f(x_1, \ldots, x_{i-1}, X_i, x_{i+1}, \ldots, x_n)\right]$$

then

$$\mathrm{Ent}\left[f(X_1, \ldots, X_n)\right] \leq \sum_{i=1}^{n} \mathbb{E}\left[g_i(X_1, \ldots, X_n)\right]$$

Bounded Difference Inequalities

a stronger version derived by entropy method

If $\{X_i\}_{i=1}^n$ indep. $\|D_i f\|_\infty < \infty$ for all i, then

$$\|f - \mathbb{E}f\|_{\psi_2}^2 \lesssim \left\| \sum_{i=1}^n |D_i f|^2 \right\|_\infty$$

Outline

Outline

Intro - Suprema

- Previous chapters: the concentration of a function $f \to \mathbb{E}f$
- This chapter: the magnitude of the mean $\mathbb{E}f$

Suprema: focus on a class of function

$$f = \sup_{t \in T} X_t$$

and discuss the relation of $\mathbb{E}(\sup_{t \in T} X_t)$ with complexity of T.

Outline

Finite Maxima

Suppose $|T| < \infty$

- if X_t have bounded first moment,
  $\mathbb{E}\left(\sup_{t\in T} X_t\right) \leq \sum_{t\in T} \mathbb{E}|X_t| \leq |T| \cdot \sup_{t\in T} \mathbb{E}|X_t|$
- if X_t have p-th bounded moment,
  $\mathbb{E}\left(\sup_{t\in T} X_t\right) \leq \left[\mathbb{E}\left(\sup_{t\in T} |X_t|^p\right)\right]^{\frac{1}{p}} \leq |T|^{\frac{1}{p}} \cdot \sup_{t\in T}(\mathbb{E}|X_t|^p)^{\frac{1}{p}}$
- -[Maximal inequality] Suppose that $\log \mathbb{E}[e^{\lambda X_t}] \leq \psi(\lambda)$ for all $\lambda \geq 0$ and $t \in T$, where ψ is convex and $\psi(0) = \psi'(0) = 0$. Then

  $$\mathbb{E}\left(\sup_{t\in T} X_t\right) \leq (\psi^*)^{-1} \log |T|$$

  - if X_t is σ^2-subg, then

    $$\mathbb{E}\left(\sup_{t\in T} X_t\right) \leq \sqrt{2\sigma^2 \log |T|}$$

- -[Maximal tail inequality]

Application of finite maxima

Model Selection Wang 2026

Outline

The Covering Method

Let T_ε be an ε-cover of T, $\forall t \in T,\ \exists\, \pi_0(t) \in T_\varepsilon,$ s.t. $d(t, \pi_0(t)) \le \varepsilon$

$$
\begin{aligned}
\mathbb{E}\left(\sup_{t\in T} X_t\right) &\le \inf_{\varepsilon>0}\left\{\mathbb{E}\left(\sup_{t\in T} X_{\pi_0(t)}\right) + \mathbb{E}\left(\sup_{t\in T}\left|X_t - X_{\pi_0(t)}\right|\right)\right\} \\
&\le \inf_{\varepsilon>0}\left\{\mathbb{E}\left(\sup_{t\in T} X_{\pi_0(t)}\right) + \varepsilon\mathbb{E}C\right\} \quad \text{(C-Lip process)} \\
&\le \inf_{\varepsilon>0}\left\{(\psi^*)^{-1}\log|T_\varepsilon| + \varepsilon\mathbb{E}C\right\} \quad \text{(finite maxima)}
\end{aligned}
$$

How to bound $\mathbb{E}\left(\sup_{t\in T} X_t\right)$ if $|T|$ is infinite? Discretization.

- We need the continuity of the random process $\{X_t\}_{t\in T}$. [Lipschitz]
- And find finite representatives of X_t. How large is the finite representatives? [covering number]
- so that for arbitrary $t \in T$ we can represent *sup*
- apply the finite maximal ineq. to an optimal ε-net.

Covering Number

[Covering number] Consider a metric space (T, d)

- A finite set $T_\varepsilon \subset T$ is called an ε-**net** (also called ε-cover) of T if (every point in T_ε is within distance ε of some point of T)

$$\forall t \in T,\ \exists \pi(t) \in N\ s.t.\ d(t, \pi(t)) \leq \varepsilon.$$

- The smallest cardinality of an ε-net for (T, d) is called the **covering number**. (a measure of the complexity of the set T at the scale ε)

$$\mathcal{N}(T, d, \varepsilon) := \inf\{|T_\varepsilon|\}.$$

[Metric entropy] $\log \mathcal{N}(T, d, \varepsilon)$

[Covering number of unit ball] $B_d(x, r) := \left\{y \in \mathbb{R}^d : \|y - x\|_2 \leq r\right\}$.
Then,

$$\left(\frac{1}{\varepsilon}\right)^d \leq \mathcal{N}\left(B_d(0,1), \|\cdot\|_2, \varepsilon\right) \leq \left(1 + \frac{2}{\varepsilon}\right)^d, \quad \forall \varepsilon > 0.$$

Continuity of a random process

Lipschitz process

[Lipschitz process] The random process $\{X_t\}_{t\in T}$ is called C-Lipschitz for a metric d on T if there exists a random variable C such that

$$|X_t - X_s| \leq Cd(t,s) \quad \text{for all } t,s \in T.$$

Example:

- Lipschtiz models
- Linear class

Lipschitz Maximal Inequality

[Lipschitz Maximal Inequality]
Suppose that $\{X_t\}_{t\in T}$ is C-Lip and $\log \mathbb{E}[e^{\lambda X_t}] \le \psi(\lambda)$ for all $\lambda \ge 0$ and $t \in T$, where ψ is convex and $\psi(0) = \psi'(0) = 0$. Then,

$$\mathbb{E}\left(\sup_{t\in T} X_t\right) \le \inf_{\varepsilon>0}\left\{(\psi^*)^{-1}\left[\log \mathcal{N}(T,d,\varepsilon)\right] + \varepsilon \cdot \mathbb{E}C\right\}.$$

- if X_t is σ^2-subg, then

$$\mathbb{E}\left(\sup_{t\in T} X_t\right) \le \inf_{\varepsilon>0}\left\{\sqrt{2\sigma^2\left[\log \mathcal{N}(T,d,\varepsilon)\right]} + \varepsilon \cdot \mathbb{E}C\right\}.$$

Outline

The Chaining Method

How to bound $\mathbb{E}\left(\sup_{t \in T} X_t\right)$ if $|T|$ is infinite? And the random process $\{X_t\}_{t \in T}$ is Lipschitz "in probabaility"?
The one-resolution discretization may give us sub-optimal bounds of the generalization gap.

Continuity of a random process

ψ process, separable process

Dudley

multi-resolution approximation

[Dudley] Let $\{X_t\}_{t\in T}$ be a separable Ψ-proc. on a metric space (T, d). Then,

$$\mathbb{E}\left(\sup_{t\in T} X_t\right) \leq 3 \times \sum_{k\in\mathbb{Z}} 2^{-k} \cdot (\psi^*)^{-1}\left(2\log N(T, d, 2^{-k})\right)$$

(Dudley's integral inequality)

Outline

Intro - Learning Theory

Recall classic statistics: parameter estimation (e.g. estimate the mean of a certain distribution)

- data: $\{x_i\}_{i=1}^n$
- hypothesis: distribution model of X
- method: statistics (e.g. $\hat{\theta} = \bar{X}$)
- analyze/evaluate: properties of the statistics ($\mathbb{E}(\hat{\theta} - \theta)$, ...) with n/?

Now in supervised learning: function estimation, i.e. parameter estimation under fixed functional form (e.g.)

- data: $\{(x_i, y_i)\}_{i=1}^n$
- hypothesis: distribution model of (X, Y), $f_\theta : \mathcal{X} \to \mathcal{Y}, \ \theta \in \mathbb{R}^m$
- method: empirical risk minimization

$$\widehat{\mathcal{R}}_n(\theta) = \frac{1}{n} \sum_{i=1}^n \ell(f_\theta(x_i), y_i), \quad \hat{\theta} = \arg\min_\theta \widehat{\mathcal{R}}_n(\theta)$$

- analyze/evaluate: population risk

$$\mathcal{R}(\theta) = \mathbb{E}[\ell(f_\theta(X), Y)].$$

with n, m, T

More on Learning theory, see `xx`

Here we only discuss supervised learning, more

Measurements of a function estimation

How well a model approximates the distribution of pair r.v.

Let's first consider the question:
Suppose we are given (1) the (empirical or population) distribution of a two-dimension pair r.v. $z = (x, y)$, (2) a loss function $\ell : \mathcal{F} \times \mathcal{Z} \to \mathbb{R}_{\geq 0}$, $\ell(f, z) := \ell(f(x), y)$, how to qualify that how well a model $f : \mathcal{X} \to \mathcal{Y}$ approximates the distribution of the pair r.v. z

- empirical risk

$$\hat{\mathcal{R}}(f) = \frac{1}{n} \sum_{i=1}^{n} \ell(f, z_i)$$

- population risk

$$\mathcal{R}(f) = \mathbb{E}_{z \sim \mu}[\ell(f, z)]$$

Examples of loss function

- Mean Squared Loss $\ell(f, (x, y)) = (f(x) - y)^2$
- cross entropy $\ell(f, (x, y)) = -y \log(f(x)) - (1 - y) \log(1 - f(x))$

Note that the population risk equals $\mathbb{E}[\hat{\mathcal{R}}(f)]$ if we consider z to be an $n * 2$-dimension r.v.[?]

Measurements of a function estimation

How well f approximates f^*

Suppose we are more given (3) a target model $f^* : \mathcal{X} \to \mathcal{Y}$, how to qualify how well f approximates f^*?

- excess risk

$$R(f) - R(f^*) = \mathbb{E}_{z\sim\mu}[l(f, z) - l(f^*, z)]$$

generalization error

$$\mathcal{E}(f; f^*) := \mathbb{E}_{z\sim\mu}[\ell(\hat{f}(x), f^*(x))].$$

Excess risk under some assumption

Note that definitions above is without assumption(we did not specify any distribution or certain function), now lets consider the question of
estimate a function within a function class
Suppose now we only consider function that is in a function class $\mathcal{F}$

Gaussian Complexity

$$\underbrace{R(f) - R(f^*)}_{\text{excess risk}} = \underbrace{\frac{1}{n}\sum_{i=1}^{n} |f(x_i) - f^*(x_i)|^2}_{\text{define this to be } \|f - f^*\|_n^2}$$

proof of Efron-Stein

Tensorization, martingale method

$\mathrm{Var}(f) = \mathbb{E}[(\underline{\underline{f - \mathbb{E}f}})^2]$

$\mathbb{E}f \to \mathbb{E}(f \mid X_1) \to \mathbb{E}(f \mid X_1, X_2) \to \cdots \mathbb{E}(f \mid X_1, \dots, X_n)$

$\underbrace{}_{\Delta_1} \quad \underbrace{}_{\Delta_2} \quad \cdots \quad = f$

Let $\Delta_k = \mathbb{E}(f \mid X_1, \dots, X_k) - \mathbb{E}(f \mid X_1, \dots, X_{k-1})$ martingale increments

$\underline{\underline{f - \mathbb{E}f}} = \sum_{k=1}^{n} \Delta_k$

so $\mathrm{Var}(f) = \mathbb{E}[(f - \mathbb{E}f)^2] = \mathbb{E}[(\sum_{k=1}^{n} \Delta_k)^2]$

$= \sum_{k=1}^{n} \mathbb{E}[\Delta_k^2]$ $(\mathbb{E}(\Delta_k \Delta_l) = 0)$ ①

And $\Delta_k = \mathbb{E}(f \mid X_1, \dots, X_k) - \mathbb{E}(f \mid X_1, \dots, X_{k-1})$

$= \mathbb{E}[(f - \mathbb{E}(f \mid X_{-k})) \mid X_1, \dots, X_k]$

$\Delta_k^2 \overset{\text{Jensen}}{\leq} \mathbb{E}[(f - \mathbb{E}(f \mid X_{-k}))^2 \mid X_1, \dots, X_k]$

$\mathbb{E}[\Delta_k^2] \leq \mathbb{E}[(f - \mathbb{E}(f \mid X_{-k}))^2]$

$= \mathbb{E}[\mathbb{E}[(f - \mathbb{E}(f \mid X_{-k}))^2 \mid X_{-k}]]$

$= \mathbb{E}[\mathrm{Var}(f \mid X_{-k})]$ ②

① + ② $\Rightarrow$ $\mathrm{Var}(f) \leq \sum_{k=1}^{n} \mathbb{E}[\mathrm{Var}(f \mid X_{-k})]$

proof of Hoeffding

Assumption: r.v. X takes values in $[a, b]$

Let $\psi(\lambda) = \log \mathbb{E} e^{\lambda(X - \mathbb{E}X)}$

NTS: $\psi(\lambda) \le \dfrac{\lambda^2 (a-b)^2}{8}, \quad \forall \lambda \in \mathbb{R}$

Taylor expansion

$$\psi(\lambda) = \psi(0) + \psi'(0)(\lambda - 0) + \frac{\psi''(\xi)}{2}(\lambda - 0)^2 \quad \xi \in [0, \lambda]$$

→ bound $\psi(0)$, $\psi'(0)$, $\psi''(\lambda)$

WLOG, assume $\mathbb{E}X = 0 \Rightarrow \begin{cases} \psi(\lambda) = \log \mathbb{E} e^{\lambda X} \\ \psi(0) = 0 \end{cases}$

$$\psi'(\lambda) = \frac{\mathbb{E}(X e^{\lambda X})}{\mathbb{E}(e^{\lambda X})} \qquad \psi'(0) = \mathbb{E}X = 0$$

$$\psi''(\lambda) = \frac{\mathbb{E}(X^2 e^{\lambda X})\mathbb{E}(e^{\lambda X}) - \mathbb{E}(X e^{\lambda X})\mathbb{E}(X e^{\lambda X})}{(\mathbb{E}(e^{\lambda X}))^2}$$

$$= \frac{\mathbb{E}(X^2 e^{\lambda X})}{\mathbb{E}(e^{\lambda X})} - \left(\frac{\mathbb{E}(X e^{\lambda X})}{\mathbb{E}(e^{\lambda X})}\right)^2$$

exponential reweighted variance

Define Q through $\dfrac{dQ}{dP} = \dfrac{e^{\lambda X}}{\mathbb{E}(e^{\lambda X})}$

$$\psi''(\lambda) = \mathbb{E}_Q X^2 - (\mathbb{E}_Q X)^2 = \mathrm{Var}_Q(X)$$ tilting

$$\le \frac{(a-b)^2}{4} \qquad \text{since } \mathrm{supp}_Q(X) \subseteq [a, b]$$

$$\Rightarrow \psi(\lambda) = 0 + 0 + \frac{\psi''(\xi)}{2}(\lambda - 0)^2$$

$$\le \frac{\lambda^2 (a-b)^2}{8}$$

$\Rightarrow$ X is $\dfrac{(a-b)^2}{4}$-subgaussian proof of McDiarmid

Let $\mathcal{F}_0 = \{\phi, \Omega\}$, $\mathcal{F}_i = \sigma(X_1, \dots, X_i)$

$\Delta_i = \mathbb{E}(f \mid \mathcal{F}_i) - \mathbb{E}(f \mid \mathcal{F}_{i-1})$

$\underline{f - \mathbb{E}f} = \sum_{i=1}^n \Delta_i$

NTS: $\sum_{i=1}^n \Delta_i$ is $\frac{1}{4}\sum_{i=1}^n \|D_i f\|_\infty^2$ -subg

By Azuma-Hoeffding, we can construct $a_i \le \Delta_i \le b_i$

that $|b_i - a_i|$ is bounded by $\|D_i f\|_\infty$ $\forall i$

$\Delta_i \le \mathbb{E}(\sup_{X_i} f \mid \mathcal{F}_i) - \mathbb{E}(f \mid \mathcal{F}_{i-1})$

$= \mathbb{E}(\sup_{X_i} f \mid \mathcal{F}_{i-1}) - \mathbb{E}(f \mid \mathcal{F}_{i-1})$

$= \mathbb{E}[(\sup_{X_i} f - f) \mid \mathcal{F}_{i-1}] \triangleq b_i$

$\Delta_i \geq \mathbb{E}[(f - \inf_{X_i} f) \mid \mathcal{F}_{i-1}] \triangleq a_i$

so $|b_i - a_i| = |\mathbb{E}[\sup_{X_i} f - \inf_{X_i} f) \mid \mathcal{F}_{i-1}]|$

$= |\mathbb{E}[D_i f \mid \mathcal{F}_{i-1}]| \le \|D_i f\|_\infty$

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